Role of the Vishniac Magnetic Helicity Flux in Mean-field Galactic Dynamos

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Acknowledgement

I would like to express my deepest gratitude to Luke Chamandy, my supervisor, for his guidance, patience, and continuous support throughout this project. I am also thankful to my collaborators Kandaswamy Subramanian, Sharanya Sur, and Kishore Gopalakrishnan for their valuable input and inspiring discussions. Your work laid the groundwork for mine, and I'm lucky to have learned from you. A special thanks to Frederick A. Gent for generously sharing his simulation data, which formed an important part of this thesis. Lastly, I'm grateful to my friends and family for their constant encouragement, love, and emotional support during this journey.

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Abstract

Magnetic fields are crucial to the evolution of galaxies, but the mechanisms amplifying and sustaining large-scale fields remain unclear. In disc galaxies, the evolution of large-scale magnetic fields is explained through mean-field dynamo theory. This theory focuses on the mean magnetic field ($\overline{\mathbf{B}}$), while the role of random magnetic fields ($\mathbf{b}$) is often overlooked. These smaller-scale fields serve as a helicity reservoir. Without effective fluxes to expel excess helicity, the buildup of magnetic helicity can suppress the α-effect, which is proportional to the mean kinetic helicity density. This process is known as catastrophic quenching. Helicity fluxes can counteract this by redistributing helicity. In this study, we incorporate one such helicity flux, the Vishniac flux, into numerical models for the first time, building on the analytical framework of Gopalakrishnan & Subramanian (2023). Using a numerical dynamical quenching model, we assess the Vishniac flux's role in the evolution of galactic magnetic fields. The Vishniac flux transitions from dynamo suppression to enhancement at critical ξ_0 values, ξ_0 represents the ratio of magnetic energy density to kinetic energy density in the system. We find that the Vishniac flux undergoes a sign reversal at a critical value of ξ_0, marking a transition from dynamo suppression to enhancement. At sufficiently high super-equipartition values of ξ_0, the flux becomes strong enough to independently drive magnetic field growth and lead to saturation, even in the absence of kinetic helicity. This occurs through a positive feedback loop, where the Vishniac flux generates an effective α-effect via helicity transport, bypassing the need for kinetic helicity (α_k). This reveals a novel mechanism for dynamo action, where the Vishniac flux not only prevents catastrophic quenching but also serves as a substitute driver for magnetic field amplification, purely driven by helicity transport and magnetic topology.

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1 Introduction

Magnetic fields are key in many astrophysical systems and processes, from star formation to cosmic ray propagation in the interstellar medium (ISM). This work focuses on magnetic fields in disc galaxies, where observations reveal large-scale, coherent fields with typical strengths of around 10μG. A natural first question is: where do these magnetic fields come from? Are they primordial, i.e., fossil relics from the early universe, or is there an ongoing mechanism that continuously regenerates them?

To answer this, we turn to the nature of the plasmas that host these fields. These plasmas are highly turbulent. A defining feature of turbulence is the cascade of energy to smaller scales. Thus, a primordial magnetic field would struggle to survive over long timescales in high turbulence environments, as magnetic energy drains to smaller scales; ohmic dissipation is both dominant and efficient. This suggests that some mechanisms must actively sustain and even amplify these fields. That mechanism is the hydromagnetic dynamo, which refers to the process by which kinetic energy is spontaneously converted into magnetic energy.

The evolution of magnetic fields can be very well described by the equations of single-fluid magnetohydrodynamics (MHD) in most astrophysical plasmas. In this formalism, the plasma behaves as a conducting fluid influenced by fluid and electromagnetic forces. However, solving the full MHD equations in turbulent regimes presents significant computational challenges due to the vast range of spatial and temporal scales involved. Thus, we rely on simplified models to study these fields. A well-accepted model is the mean field dynamo theory, where fields are decomposed into mean (large-scale) and fluctuating (small-scale) components, and we focus on mean-field growth. This theory relies on turbulent flows breaking mirror symmetry, enabling magnetic fields to grow at scales larger than the turbulence[2][9].

Kinetic helicity is one of the most natural measures of the lack of reflectional symmetry and it arises naturally in rotating systems. The resulting large-scale magnetic fields are characterized by magnetic helicity, which measures the twist, writhe, and linkage of field lines. Kinetic and magnetic helicity are central to understanding magnetic field amplification and evolution[3]. Here, we adopt the $\alpha\Omega$ mean field dynamo model. The α-effect converts toroidal magnetic fields into poloidal fields through the action of kinetic helicity, while the Ω-effect converts poloidal fields back into toroidal fields, forming the basis of the $\alpha\Omega$-dynamo model.

However, the amplification of large-scale magnetic fields in mean-field dynamos is limited. As large-scale fields grow, small-scale magnetic helicity of opposite sign to that of the large-scale magnetic helicity is generated to conserve total magnetic helicity. As this small-scale magnetic helicity accumulates, it reduces dynamo efficiency by suppressing the α-effect. Mechanisms to remove or redistribute small-scale helicity are crucial to maintaining dynamo activity. While resistive dissipation is too slow, processes such as large-scale outflows may expel helicity. However, in many astrophysical systems, such large outflows may not be present, which calls for alternative mechanisms to facilitate helicity transfer. Thus, the standard dynamo models may not fully capture the mechanisms driving large-scale magnetic field growth.

Vishniac (2012) [11] proposed a new effect arising from higher-order terms, known as the Vishniac flux, which is a magnetic helicity flux. The Vishniac flux arises from gradients in the random magnetic and kinetic energies along the direction of large-scale rotation. As a result, it can operate in any rotating, stratified system. This project aims to incorporate the Vishniac flux into the traditional mean-field dynamo model and assess its impact on the evolution of galactic magnetic fields. The analytic expression for the Vishniac flux, derived by Gopalakrishnan & Subramanian (2023), is integrated into a dynamical quenching model developed by Chamandy et al. (2014) [5]. Our model simultaneously solves the mean-field induction equation and the

helicity evolution equation.

Unlike traditional models that primarily rely on kinetic energy-specifically the kinetic α-effect and helicity,as the main drivers of the dynamo, our model suggests that magnetic energy, in combination with large-scale rotation or shear, could be sufficient to sustain the dynamo. However, this is only true when the small-scale field strength exceeds the turbulent kinetic energy, i.e., in a super-equipartition state where small-scale fields dominate. This perspective challenges some foundational aspects of dynamo theory and aligns with our goal of providing a more comprehensive explanation for the observed magnetic field structures in galaxies.

2 Theoretical Background

The evolution of the magnetic field in a plasma is governed by the induction equation:

$$\frac{\partial \mathbf{B}}{\partial t} = \nabla \times (\mathbf{V} \times \mathbf{B}) + \eta \nabla^2 \mathbf{B}, \tag{1}$$

where $\eta = c^2/(4\pi\sigma)$ is the magnetic diffusivity. The first term in Eq. (1) is the induction term, representing magnetic field generation by plasma motion, and the second term is the diffusion term, representing field decay due to finite conductivity.

Using Reynolds averaging, the velocity and magnetic fields are decomposed as:

$$\mathbf{V} = \overline{\mathbf{V}} + \mathbf{v}, \quad \mathbf{B} = \overline{\mathbf{B}} + \mathbf{b}, \tag{2}$$

where $\overline{\mathbf{F}}$ is the mean and $\mathbf{f}$ the fluctuating component for a field $\mathbf{F}$ with $\overline{\mathbf{v}} = \overline{\mathbf{b}} = 0$.

Applying this to Eq. (1), the mean-field induction equation becomes:

$$\frac{\partial \overline{\mathbf{B}}}{\partial t} = \nabla \times \left(\overline{\mathbf{V}} \times \overline{\mathbf{B}} + \boldsymbol{\mathcal{E}} - \eta \nabla \times \overline{\mathbf{B}} \right), \tag{3}$$

where $\boldsymbol{\mathcal{E}} = \overline{\mathbf{v} \times \mathbf{b}}$ is the mean electromotive force (EMF).

The mean EMF is expressed as:

$$\boldsymbol{\mathcal{E}} = \alpha \overline{\mathbf{B}} - \eta \nabla \times \overline{\mathbf{B}}, \tag{4}$$

where α represents helicity-driven amplification of the magnetic field (the α-effect) and β accounts for turbulent diffusivity, which opposes field growth.The α-effect works by generating electric currents through helical turbulent motions, which interact with the toroidal magnetic field. These currents create a force that "twists" the toroidal field into a poloidal field. Essentially, the motion of the fluid moves the magnetic field lines in such a way that the circular toroidal field becomes a looped poloidal field, transferring the magnetic energy from one configuration to the other. This is how the α-effect converts a toroidal field into a poloidal field. In disc galaxies, the mean velocity field $\overline{\mathbf{V}}$ is dominated by differential rotation. This rotation induces the Ω-effect. This effect involves the stretching of magnetic field lines by differential rotation. It converts poloidal fields into toroidal fields, enabling a regenerative cycle when combined with the α effect. Together, the α and Ω effects form the $\alpha\Omega$-dynamo, supporting exponential growth of the mean magnetic field ($\overline{\mathbf{B}}$) if they overcome turbulent diffusion.

The α-effect in Equation (4) is represented as the sum of kinetic and magnetic contributions:

$$\alpha = \alpha_k + \alpha_m, \tag{5}$$

where:

$$\alpha_k = -\frac{\tau}{3}\overline{\mathbf{u}\cdot(\nabla\times\mathbf{u})} \quad \text{is the kinetic part related to the mean helicity of the random flow,} \tag{6}$$

$$\alpha_m = \frac{\tau}{12\pi\rho}\overline{\mathbf{b}\cdot(\nabla\times\mathbf{b})} \quad \text{is the magnetic contribution.} \tag{7}$$

Here, ρ is the gas density, τ is the correlation time of the random flow, and $\mathbf{u}$ and $\mathbf{b}$ represent the small-scale velocity and magnetic field, respectively.

In the early stages of dynamo action, the magnetic field is weak and does not significantly influence the fluid motions. The growth of the mean magnetic field is exponential, driven by induction effects, with turbulent diffusion attempting to dissipate it. This is the kinematic regime of dynamo action. As the magnetic field grows, it starts to exert a back-reaction on the flow via the Lorentz force. This alters the dynamics of the random velocity field, reducing the effectiveness of the kinetic α_k-effect and turbulent diffusivity (β). The system enters a non-linear regime where the magnetic helicity and magnetic α_m-effect become significant, introducing feedback mechanisms. α_m and α_k have opposite signs, thus as $\overline{\mathbf{b}\cdot(\nabla\times\mathbf{b})}$ is amplified, the magnitude of the total α-effect decreases, saturating the growth of the mean magnetic field.

The dynamo non-linearity is governed by the following equation for α_m-effect:

$$\frac{\partial\alpha_m}{\partial t} = -\frac{2\eta}{l^2 B_{\text{eq}}^2}\boldsymbol{\mathcal{E}}\cdot\overline{\mathbf{B}} - \nabla\cdot\boldsymbol{\mathcal{F}}, \tag{8}$$

where l is the correlation length of the turbulence. $B_{\text{eq}} = \sqrt{4\pi\rho}\,u$ is the characteristic magnetic field strength corresponding to energy equipartition with turbulence(this is B_0 at midplane) and $\boldsymbol{\mathcal{F}}$ is the flux density of α_m, related to the flux density of the small-scale magnetic helicity density.

There are many possible contributions to $\boldsymbol{\mathcal{F}}$. In general, a flux of the form

$$\boldsymbol{\mathcal{F}} = \boldsymbol{\mathcal{F}}^{NV} + \boldsymbol{\mathcal{F}}^{A} + \boldsymbol{\mathcal{F}}^{D} \tag{9}$$

is considered. In this project, we are not including the advective flux; thus, we have diffusive flux and the Vishniac flux for helicity transfer.

2.1 The Diffusive Helicity Flux

In earlier work,the diffusive helicity flux was modeled heuristically, with the following expression for the flux density [5]:

$$\boldsymbol{\mathcal{F}}^{D} = -\kappa\nabla\alpha_m \tag{10}$$

with $\kappa \approx 0.3\eta$, obtained from numerical simulations.
In Gopalakrishnan & Subramanian (2023), the flux density was derived more rigorously by considering higher-order terms in the magnetohydrodynamic equations. This newer form of the diffusive helicity flux is based on the physical processes underlying turbulence and the evolution of magnetic helicity. The form derived is:

$$\boldsymbol{\mathcal{F}}^{D} = -\frac{7\tau}{27}\left(\langle\mathbf{v}^2\rangle + \frac{\langle\mathbf{b}^2\rangle}{4\pi\rho}\right)\nabla h^b \tag{11}$$

Here, ρ is the gas density, τ is the correlation time of the random flow, and $\mathbf{v}$ and $\mathbf{b}$ represent the small-scale velocity and magnetic field, respectively. Following Sur et al. (2007), we define the magnetic helicity density as:

$$\frac{\partial \chi}{\partial t} + \nabla \cdot F = -2E \cdot B - 2\eta j \cdot b \tag{12}$$

$$\alpha_m = \frac{1}{3}\tau \frac{1}{\lambda^2}\frac{\chi}{\rho}\left(\frac{1}{4\pi}\right) \tag{13}$$

Assuming $\chi = h^b$ and $\lambda = v\tau$, we get:

$$h^b = \frac{3\rho\lambda^2}{\tau}\alpha_m(4\pi) = \frac{3\rho\tau^2 v^2}{\tau}\alpha_m(4\pi) = 3\rho\tau v^2 \alpha_m(4\pi) \tag{14}$$

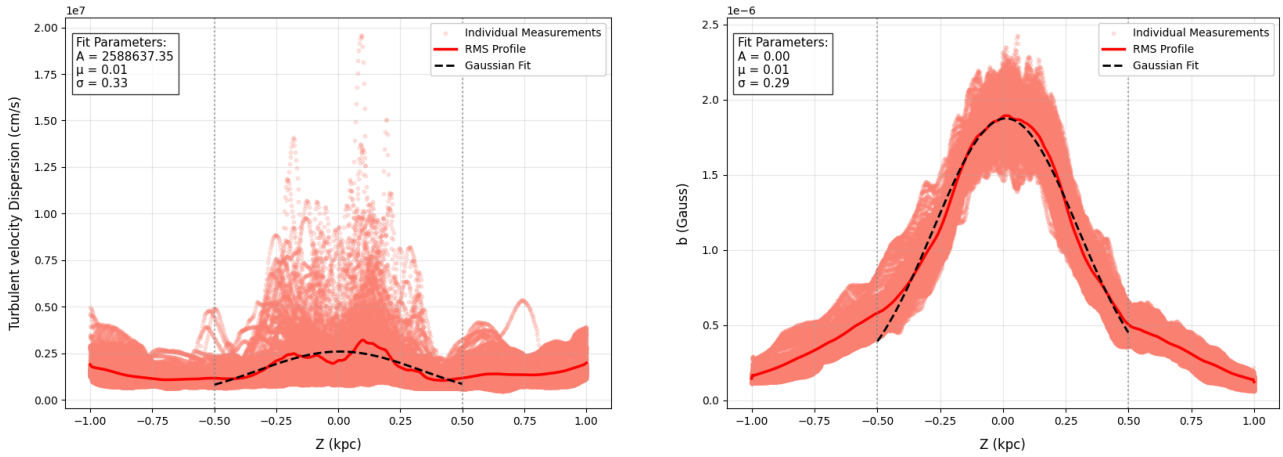


Fig. 1: Profiles of $\boldsymbol{v}$ and $\boldsymbol{b}$ from the simulation data of Gent et al. (2024)[**?**], with a gaussian fitting done.

To proceed with the analysis, we considered two distinct profiles for the turbulent velocity $\mathbf{v}$ field in our model. Given by:

$$\text{Case 1:} \quad v^2(z) = v_0^2 \exp\left(\frac{z^2}{h^2}\right), \qquad \text{Case 2:} \quad v^2(z) = v_0^2 \exp\left(-\frac{z^2}{h^2}\right), \tag{15}$$

where v_0 is a characteristic turbulent velocity amplitude and h is the disc scale height. Case 1 is motivated by Gressel et. al[8, see p. 123] and [9]. Case 2 is motivated by the simulation data of Gent et al. (2024). Through private correspondence, simulation data was obtained from Dr. Gent, including profiles of small-scale magnetic field strength and turbulent velocity across the galactic disc. These datasets were analyzed to extract the spatial structure of the turbulence and magnetic energy. To quantitatively characterize the turbulence profiles, a best-fit Gaussian model was applied to both the small-scale magnetic field and the velocity data, which is plotted in figure 1. For both cases, the profile of $b^2(z)$ is:

$$b^2(z) = b_0^2 \exp\left(\frac{z^2}{h^2}\right) \tag{16}$$

For consistency with the assumptions made in theoretical derivation of the fluxes[7], we keep the

mass density ρ constant throughout the simulations. This assumption simplifies the analysis and allows us to focus on the behavior of the turbulent velocity and magnetic fields, which are the primary variables driving the helicity flux we are interested in.
To express the ratio of magnetic to kinetic turbulent energies, we define the parameter:

$$\xi(z) = \frac{b^2(z)}{4\pi\rho v^2(z)}. \tag{17}$$

This is written as:

$$\xi(z) = \tilde{\xi}(z)\,\xi_0, \tag{18}$$

where $\xi_0 = \frac{b_0^2}{4\pi\rho v_0^2}$ is a constant. In Case 2, where both $v^2(z)$ and $b^2(z)$ have identical profiles, $\tilde{\xi}(z) = 1$, and hence $\xi(z) = \xi_0$.

Here, we consider a one-dimensional setup in the vertical direction z. Using the expression for the diffusive helicity flux derived in [7], and substituting our profiles, the divergence of the diffusive helicity flux $-\nabla \cdot \boldsymbol{\mathcal{F}}^D$ becomes:

Case 1:

$$\begin{aligned} -\nabla \cdot \boldsymbol{\mathcal{F}}^D = \frac{7\tau v_0^2}{27}\Bigg\{ &\left[\exp\left(\frac{z^2}{h^2}\right) + \xi_0 \exp\left(-\frac{z^2}{h^2}\right)\right]\frac{\partial^2 h^b}{\partial z^2} \\ &+ \frac{2z}{h^2}\left[\exp\left(\frac{z^2}{h^2}\right) - \xi_0 \exp\left(-\frac{z^2}{h^2}\right)\right]\frac{\partial h^b}{\partial z}\Bigg\} \end{aligned} \tag{19}$$

Case 2

$$-\nabla \cdot \boldsymbol{\mathcal{F}}^D = \frac{7\tau v_0^2}{27}(1+\xi_0)\exp\left(-\frac{z^2}{h^2}\right)\left[\frac{\partial^2 h^b}{\partial z^2} - \frac{2z}{h^2}\frac{\partial h^b}{\partial z}\right] \tag{20}$$

2.2 The New Vishniac Flux

$\boldsymbol{\mathcal{F}}^{NV}$ is the New Vishniac flux, not to be confused with the Vishniac-Cho flux(Vishniac & Cho 2001). Henceforth, the term "Vishniac flux" will refer to the New Vishniac flux, this flux is the main focus of the project. This is derived by Gopalakrishnan & Subramanian (2023) and is expressed as:

$$\boldsymbol{\mathcal{F}}^{NV} = (\nabla \times \overline{\mathbf{V}}) \cdot \left[C_1 \frac{\tau}{8\pi\rho}(\langle \mathbf{b}^2\rangle)^2 + C_2\tau\langle \mathbf{v}^2\rangle\langle \mathbf{b}^2\rangle + C_4\lambda^2\langle \mathbf{b}^2\rangle\right]\hat{z} + \tau(C_3 - C_2)(\overline{\mathbf{V}} \times \langle \mathbf{v}^2\rangle\nabla\langle \mathbf{b}^2\rangle) \tag{21}$$

This expression simplifies under the assumption that all turbulent correlators predominantly vary in a single spatial direction, which is physically justified in systems like disk galaxies (with vertical stratification). Thus when we take a divergence of this flux, the second term will vanish. Thus, we have:

$$-\nabla \cdot \boldsymbol{\mathcal{F}}^{NV} = -(\nabla \times \overline{\mathbf{V}}) \cdot \left[C_1 \frac{\langle b^2\rangle}{4\pi\rho}\tau\nabla\langle b^2\rangle + C_2\langle b^2\rangle\tau\nabla\langle v^2\rangle + C_3\langle v^2\rangle\tau\nabla\langle b^2\rangle + C_4\lambda^2\nabla\langle b^2\rangle\right] \tag{22}$$

$$\text{where } C_1 = \frac{7}{45}, \quad C_2 = -\frac{203}{5400}, \quad C_3 = \frac{403}{8100}, \quad \text{and} \quad C_4 = -\frac{1}{6}$$

Here, λ is the characteristic length scale of turbulence. We now compute the curl of the mean velocity field $\overline{\mathbf{V}}$. In cylindrical coordinates, this reads:

$$\nabla \times \overline{\mathbf{V}} = \left(-r\frac{\partial \Omega}{\partial z} + \frac{1}{r}\frac{\partial V_z}{\partial \phi}\right)\hat{r} - \frac{\partial V_z}{\partial r}\hat{\phi} + (2-q)\Omega\hat{z} \tag{23}$$

Here, q is the shear parameter defined by:

$$q = -\frac{d\ln\Omega}{d\ln r} \tag{24}$$

Assuming axisymmetry and negligible vertical shear ($\partial\Omega/\partial z = 0$), the expression reduces to:

$$\nabla \times \overline{\mathbf{V}} = (2-q)\Omega\hat{z} \tag{25}$$

Now, after algebraic manipulations, we will get the final expression of the Vishniac flux for case 1 and 2:

Case 1:

$$-\nabla \cdot \boldsymbol{\mathcal{F}}^{NV} = (2-q)\frac{8\pi\rho R_\Omega \eta\tau v_0^4 \xi_0}{q}\left(\frac{z}{h^4}\right)\left[0.079 - 0.156\xi_0 e^{-2z^2/h^2}\right] \tag{26}$$

Case 2:

$$-\nabla \cdot \boldsymbol{\mathcal{F}}^{NV} = (2-q)\frac{8\pi\rho R_\Omega \eta\tau v^4 \xi_0}{q}\left(\frac{z}{h^4}\right)\left[0.154 - 0.156\xi_0\right] e^{-2z^2/h^2} \tag{27}$$

We introduced the dimensionless parameter R_Ω here, which will control Ω by the relation:

$$\Omega = \frac{-R_\Omega \eta}{qh^2} \tag{28}$$

Here, η is the turbulent diffusivity given by:

$$\eta = \frac{1}{3}\tau v^2 \tag{29}$$

3 The model

3.1 Assumptions

We adopt a cylindrical coordinate system (r, ϕ, z) and solve the induction equation (Equation 3) under the assumption of axisymmetry, which neglects azimuthal (ϕ) derivatives. Applying the thin-disc approximation, the vertical magnetic field component is considered negligible compared to the radial and azimuthal components, i.e., $|\overline{B}_z| \ll |\overline{B}_r|, |\overline{B}_\phi|$, and radial ($r$) derivatives are neglected. These simplifications reduce the problem to a one-dimensional form in z, allowing local dynamo solutions. We solve for $\overline{B}_r$, $\overline{B}_\phi$, and α_m. Using $\nabla \cdot \overline{\mathbf{B}} = 0$, the vertical magnetic field $\overline{B_z}$ is determined.

The reduced induction equations for the $\alpha\Omega$ dynamo are:

$$\frac{\partial \overline{B}_r}{\partial t} = -\frac{\partial}{\partial z}\left(\alpha\overline{B}_\phi\right) + \eta\frac{\partial^2 \overline{B}_r}{\partial z^2} \tag{30}$$

$$\frac{\partial \overline{B}_\phi}{\partial t} = -\Omega \overline{B}_r + \eta \frac{\partial^2 \overline{B}_\phi}{\partial z^2} \tag{31}$$

The α_m evolution equation given in equation 8 becomes:

$$\frac{\partial \alpha_m}{\partial t} = -\frac{2\eta}{l^2 B_{\rm eq}^2}\left[\alpha\left(\overline{B}_r^2 + \overline{B}_\phi^2\right) - \eta\left(\frac{\partial \overline{B}_r}{\partial z}\overline{B}_\phi - \frac{\partial \overline{B}_\phi}{\partial z}\overline{B}_r\right)\right] - \nabla\cdot\boldsymbol{\mathcal{F}}^{NV} - \nabla\cdot\boldsymbol{\mathcal{F}}^{D}, \tag{32}$$

The magnetic pitch angle is the angle between the azimuthal and radial components of the magnetic field in a disk, indicating the spiral structure of the field:

$$p = \arctan\left(\frac{\overline{B}_r}{\overline{B}_\phi}\right), \quad p \in \left(-\frac{\pi}{2}, \frac{\pi}{2}\right) \tag{33}$$

In a thin disc, the α-effect is written as:

$$\alpha_k = \alpha_0(r)\tilde{\alpha}(z), \tag{34}$$

where $\alpha_0 = \frac{l^2\Omega}{h}$ with h as the disc half-thickness. From symmetry considerations, α is an odd function of z, a conventional choice of the function is[5]:

$$\tilde{\alpha}(z) = \sin\left(\frac{\pi z}{h}\right). \tag{35}$$

We consider specific models at $r = 4\,\mathrm{kpc}$. Dimensionless control parameters are defined as:

$$R_\alpha \equiv \frac{\alpha_0 h}{\eta}, \quad R_\Omega \equiv \frac{-q\Omega h^2}{\eta}, \quad R_\kappa \equiv \frac{\kappa}{\eta} \tag{36}$$

Lengths are scaled to the disk half-thickness h, except for radial measurements, which are scaled to the disk radius R. The magnetic field is scaled to $B_0 \sim 1\,\mu\mathrm{G}$.

3.2 Numerical Methods

The equations (30), (31), and (32) are solved numerically using a simulation code in FORTRAN. The numerical schemes used in this project are inspired by the well-established Pencil Code, though the implementation itself has been developed independently from scratch.

3.3 Initial and Boundary Conditions

The choice of initial conditions is not critical in kinematic mean-field dynamo simulations, as the system tends to forget its initial state due to exponential growth (or decay) of magnetic modes. Here, a seed field with a Gaussian profile of amplitude 10^{-2} is used.

For boundary conditions, the standard vacuum boundary conditions are employed, where the region outside the dynamo-active disk is assumed to be a perfect electrical insulator.

$$\overline{B}_r = \overline{B}_\phi = 0, \quad \alpha_m = 0, \quad \text{at } z = \pm h. \tag{37}$$

Physically, this implies that no electric currents exist outside the disk, and hence, only the meridional components of the magnetic field (B_r and B_z) can penetrate the boundary, while the azimuthal component B_ϕ must vanish at the edge. At first glance, this assumption may

seem questionable - after all, real galactic disks are enveloped by a halo of ionized gas which does conduct. However, this ionized corona is only weakly conducting compared to the dense interstellar medium in the disk, and it behaves like a turbulent electric vacuum for the purposes of modeling large-scale fields inside the disk.

3.4 Implementation

Spatial derivatives are approximated using a sixth-order finite differencing method, which balances accuracy and efficiency. Higher-order schemes (8th and 12th) showed only marginal gains in accuracy but were more computationally demanding, requiring increased memory and longer run times. Hence, the sixth-order method was chosen as the most efficient solution.

Time evolution is handled using a third-order implicit Runge–Kutta (RK3) scheme [4]. Although a fourth-order explicit RK4 scheme was also tested, no significant difference in the results was observed. A key advantage of the implicit RK3 over the explicit RK4 scheme is its stability when handling the stiff equations that arise in MHD simulations, where processes like diffusion and advection operate on different timescales. The implicit RK3 scheme enables larger time steps while maintaining stability, making it more computationally efficient than the RK4 scheme, which requires significantly smaller time steps, thereby increasing memory usage and run time to maintain stability.

The simulation employs a spatial resolution of 101 grid points. While models were also tested with 51 and 201 grid points, the 201-grid resolution, though providing slightly more accurate results, was significantly more time-consuming. Therefore, a resolution of 101 grid points was used, and every new model was also tested with 201 grid points to see if there are any pronounced differences.

Boundary conditions are implemented using ghost zones, which are additional grid points beyond the physical domain of the simulation. These ghost zones are not part of the actual solution, but are used to impose boundary conditions by mirroring, extrapolating, or applying specific rules to the data inside the physical grid. In the code, three ghost cells are placed on each side of the physical grid, resulting in a total of 107 grid points. The use of ghost zones is important for maintaining the accuracy of the finite differencing scheme near the boundaries, as they allow for consistent calculation of derivatives and prevent numerical instabilities that might arise from directly enforcing boundary conditions.

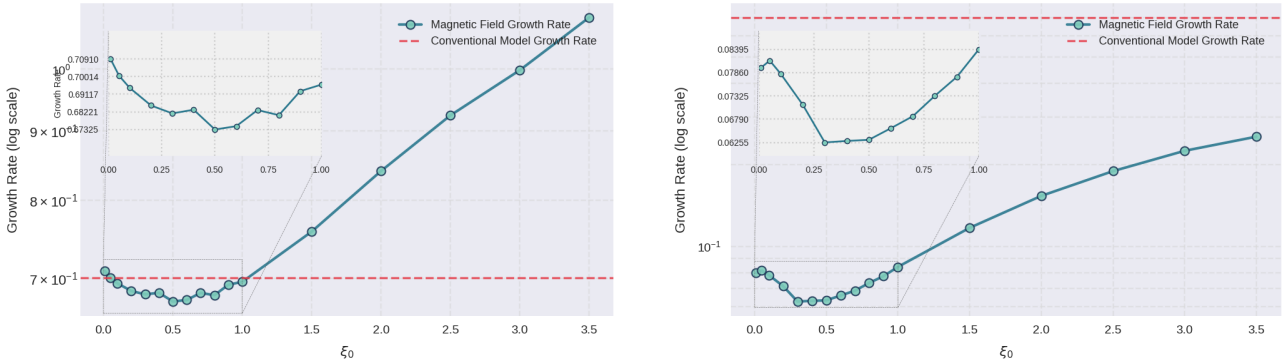
4 Results and Analysis

4.1 The Vishniac Flux in Mean-Field Dynamos: Aid or Hindrance?

We carried out a parameter sweep over ξ_0, keeping $v_0 = 20$ and $R_\Omega = -20$. This was done to assess whether the Vishniac flux helps or hinders the dynamo action, and to check if there's a critical window of ξ_0 values where it begins to help. The plots of growth rate vs ξ_0 and saturation strength vs ξ_0 for both Case 1 and Case 2 are shown in the figures.

We varied ξ_0 from 0.01 to 3.5. Although such large values of ξ_0 are not realistic in astrophysical settings, we included them intentionally to explore the theoretical behavior of the model. In reality, astrophysical ξ_0 is expected to be much lower, but it's at these large values that the Vishniac flux starts significantly boosting the dynamo action.

In Figure 2, we show how the growth rate changes as ξ_0 varies from 0.01 to 3.5. The red line



(a) Case 1

(b) Case 2

Fig. 2: Growth rate values for Case 1 and Case2 across different ξ_0 values. All tested models include the kinetic α-effect (α_k). Case 1 and Case 2 correspond to the different velocity profiles given in 15

shows the growth rate from the conventional model without the Vishniac term. Looking at Case 1, it's clear that below a certain critical ξ_0, the Vishniac flux works against the dynamo by reducing the growth rate. This is consistent with the derived expression for the Vishniac flux in Case 1:

$$-\nabla \cdot \boldsymbol{\mathcal{F}}^{NV} \propto \left[0.079 - 0.156\xi_0 e^{-2z^2/h^2}\right] \tag{38}$$

From this, we can see that there's a critical value of ξ_0 where the sign of the flux changes. This value can be calculated numerically and comes out to be $\xi_0 = 0.506$. In Figure 2a, we can clearly see the effect of this transition point. For $\xi_0 < 0.506$, the flux is negative and contributes negatively to the evolution of α_m. It essentially fights against the driving term provided by α_k. In simulations without α_k and for $\xi_0 < 0.506$, the magnetic field decays, and the decay rate slows as ξ_0 approaches 0.506. This trend matches what we expect from the midplane analysis.

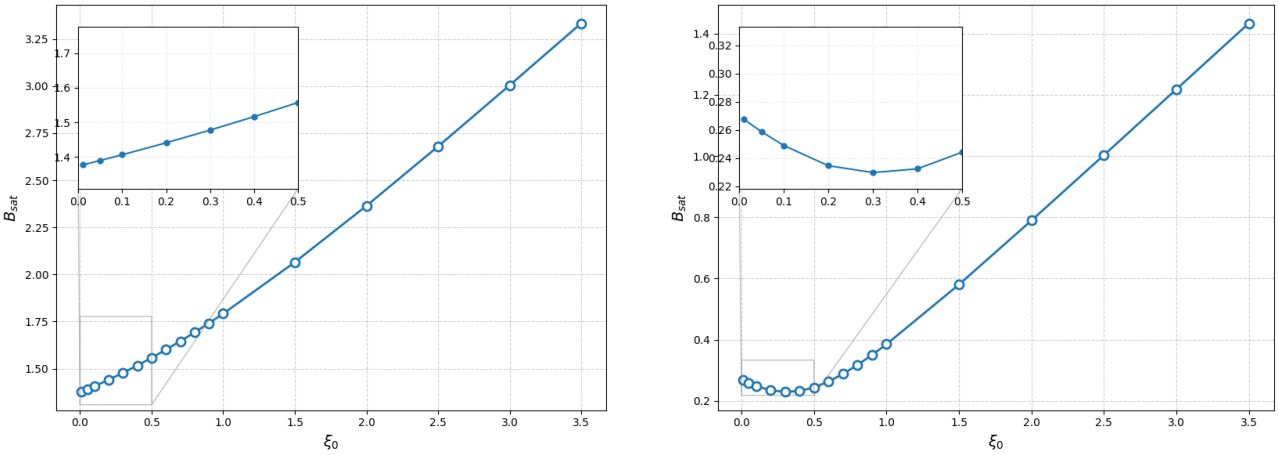
Once $\xi_0 > 0.506$, the flux flips sign and starts to help the dynamo. This is also seen in the sharp increase in growth rate past $\xi_0 = 0.5$ in Figure 2a. For $\xi_0 > 1$, where the small-scale fields become superequipartition and start dominating, the flux enhances the growth rate beyond that of the conventional dynamo. This suggests that at large enough ξ_0, the Vishniac term could possibly dominate the helicity evolution and drive the dynamo by itself. This is discussed more in the next section.

Now coming to Case 2, the flux takes the form:

$$-\nabla \cdot \boldsymbol{\mathcal{F}}^{NV} \propto \left[0.154 - 0.156\xi_0\right] e^{-2z^2/h^2} \tag{39}$$

Here, the saddle point where the flux flips sign is at $\xi_0 = 0.987$. There is also an additional exponential factor outside the bracket in Case 2, which makes the flux look like a bell curve peaking at the midplane. This means the Vishniac term has a very localized effect in this case, most of the action happens near the midplane.

From simulations (see Figure 2b), we observe that the overall trend in Case 2 is similar to Case 1. However, the increase in growth rate with increasing ξ_0 is slower. Even at $\xi_0 = 3.5$, the growth rate has not yet surpassed that of the conventional dynamo. This tells us that unless ξ_0 becomes very large, the flux remains too localized to be significantly helpful. For it to become effective, the value of ξ_0 must be high enough for the flux's influence to extend beyond just the midplane.

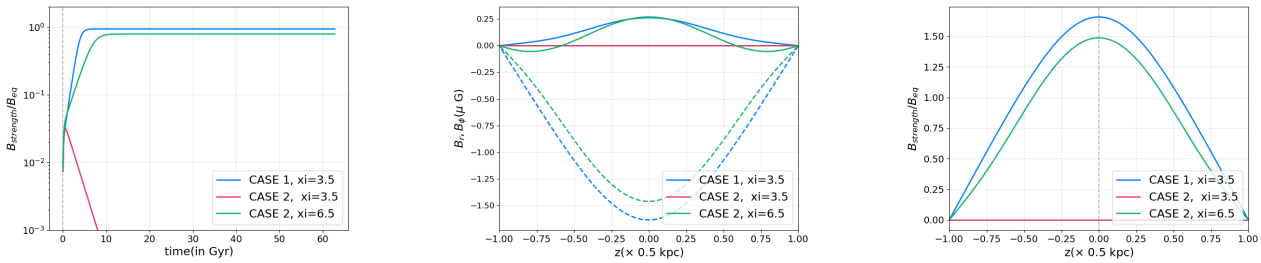


(a) Case 1

(b) Case 2

Fig. 3: Field saturation strength for Case 1 and Case2 across different ξ_0 values. All tested models include the kinetic α-effect (α_k).

4.2 The Vishniac flux action without α_k driving the dynamo



(a) *Mean field strength vs time plots, the y axis here is in logarithmic scale.*

(b) *Mean field strength vs z plots, dashed line represents $\overline{B}_\phi$ and solid lines are $\overline{B}_r$.*

(c) *Total field strength vs z plots. This is the vector sum of $\overline{B}_\phi$ and $\overline{B}_r$.*

Fig. 4: Dynamo models with pure magnetic α-effect ($\alpha = \alpha_m$), excluding the conventional kinetic contribution (α_k). Case 1 and Case 2 correspond to the different velocity profiles given in 15

As shown in the previous section, there is evidence that at superequipartition values of ξ_0, the Vishniac flux alone can sustain dynamo action. To investigate this possibility, we considered three models. Two of these are Case 1 and Case 2 evaluated with $\xi_0 = 3.5$. In both, we explicitly turned off the kinetic helicity term α_k, meaning the usual driver of large-scale field growth was absent. Surprisingly, in Case 1, we still observed a growing and eventually saturated magnetic field—driven purely by the Vishniac flux. In Case 2, the field initially decayed for the same ξ_0 value. However, based on the trend seen in Figure 2b, we can assume that even Case 2 could support dynamo action if ξ_0 were increased further. This was confirmed: at higher ξ_0, Case 2 also exhibited self-sustained field growth.

But how is this possible, given that $\alpha_k = 0$? In the conventional picture of a galactic dynamo, turbulence (e.g., from supernova explosions) produces kinetic helicity, breaking reflectional symmetry and giving rise to α_k. This, in combination with differential rotation (Ω), forms the classic $\alpha\Omega$ dynamo loop: poloidal fields generate toroidal fields and vice versa, leading to exponential field growth. When α_k is switched off, that feedback loop is broken, and we expect no sustained dynamo.

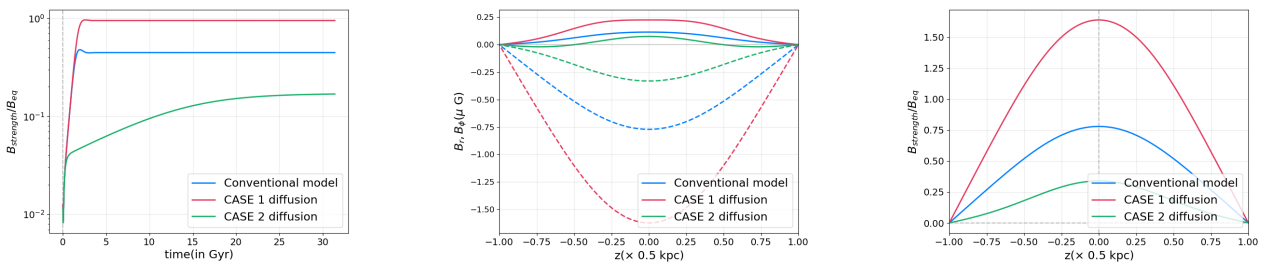
However, the Vishniac flux introduces a twist to this story. In the standard scenario, the role of the flux is to avoid catastrophic α-quenching by transporting small-scale magnetic helicity out of the dynamo-active region. But at sufficiently large ξ_0, this flux doesn't only work to remove excess helicity, but also behave as the driving term of magnetic helicity evolution equation.

This can be explained using the evolution equation for magnetic α, assume a high ξ_0 case where the flux have overall positive sign and is larger than diffusion:

$$\frac{\partial \alpha_m}{\partial t} \approx -\frac{2\eta}{l^2 B_{\rm eq}^2}\left[\alpha\left(\overline{B}_r^2+\overline{B}_\phi^2\right)-\eta\left(\frac{\partial \overline{B}_r}{\partial z}\overline{B}_\phi-\frac{\partial \overline{B}_\phi}{\partial z}\overline{B}_r\right)\right]+\boldsymbol{\mathcal{F}}^{\rm NV}. \tag{40}$$

In the early stages, when mean fields are still weak, the non-linear quenching and gradient terms are negligible. Thus, the equation 8 will not be activated yet, and α_k and Ω drive the equations 30 and 31. Once the mean fields start growing, the equation 8 begins to affect the system. Now, no α_k means no mean field. However, since the Vishniac flux can exist without any mean fields, the equation 8 will still work to evolve the system from the initial growth phase. Thus, at a large and positive Vishniac flux, the term $\boldsymbol{\mathcal{F}}^{\rm NV}$ directly activates α_m. That is, the helicity transport creates a net gradient in magnetic helicity, mimicking the role of α_k. Therefore, the Vishniac flux-generated α_m results in a non-zero total α-effect, which, together with Ω, forms the positive feedback loop required for classic mean-field dynamo action.
In short, a sufficiently strong Vishniac flux seeds an effective α-effect which is completely magnetic in origin, which enables the classic dynamo feedback loop without needing kinetic helicity. This means a dynamo is possible that is purely driven by magnetic energy.

4.3 Diffusive Flux Behavior



(a) *Mean field strength vs time plots, the y axis here is in logarithmic scale.*

(b) *Mean field strength vs z plots, dashed line represents $\overline{B}_\phi$ and solid lines are $\overline{B}_r$.*

(c) *Total field strength vs z plots. This is the vector sum of $\overline{B}_\phi$ and $\overline{B}_r$.*

Fig. 5: Comparison of conventional and modified diffusive fluxes for Cases 1 and 2 velocity profiles. Case 1 and Case 2 correspond to the different velocity profiles given in 15. The blue curve represents the conventional $\alpha\Omega$ model with the standard diffusive flux formulation. The diffusive flux strength is governed by the parameter ξ_0, which is fixed at $\xi_0 = 0.6$. There is no Vishniac flux in these models because we want to isolate the diffusion alone and analyze.

The diffusive fluxes derived by assuming case 1 and case 2 is studied in figure 5. To better understand the influence of our modified diffusive helicity flux on dynamo growth, we examine the flux expression in Case 1:

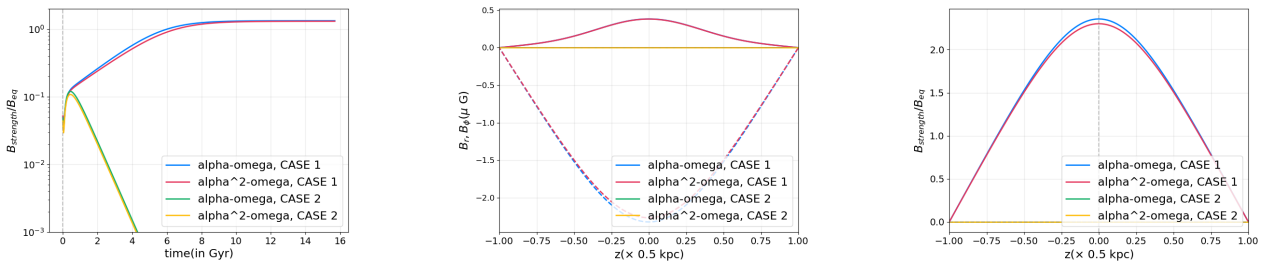
$$-\nabla\cdot\mathcal{F}^D \propto \left[\underbrace{e^{+z^2/h^2}}_{\text{tries to spread outwards}} + \underbrace{\xi_0 e^{-z^2/h^2}}_{\text{tries to contain in midplane}}\right]\frac{\partial^2 h^b}{\partial z^2} \tag{41}$$

This structure reveals a clear competition between two opposing effects: the first term tends to drive helicity outward toward the disk boundaries, while the second term proportional to the parameter ξ_0-attempts to confine helicity within the midplane. We tested this new form of diffusive flux against the considered standard model[5].

The comparison of magnetic field growth (see Figure 5) shows that the growth rates, represented by the slope during the initial exponential phase, are nearly identical for Case 1 and the conventional model. These two models both adopt the same turbulent velocity stratification. The plots given in figure 5 is from the galactic midplane. Since $z = 0$ here, the second term in equation 19 vanishes. Now the heuristic equation used in conventional model and Case 1 diffusion is effectively the same. Both proportional to second derivative of α_m, and they only differ by the constant magnitude, which in Case 1 is now monotonously dependent on ξ_0. This explains why conventional and Case 1 diffusion have same growth rate but slighly different but apparent saturation strength.
In Case 2, however, the flux expression lacks any such competition between terms. Here, the entire expression works in favor of confining helicity near the midplane. As a result diffusion is particularly strong in this region, which suppresses field growth. This is reflected in the lower growth rate observed in the simulations for Case 2. Also in Case 2, the saturation field strength is significantly lower, probably due to excessive midplane diffusion weakening the dynamo.
For self consistency, we will be using the new diffusion form, given in equations 19 and 20 when we incluse Vishniac flux in the model, since both these fluxes are derived using the same technique by Gopalkrishnan and Subramanian[7].

4.4 Comparison of the $\alpha\Omega$ vs $\alpha^2\Omega$ Dynamo Mechanisms

To examine whether the characteristics of the flux change under a different dynamo regime, we replaced the $\alpha\Omega$ model with an α^2-Ω configuration. The corresponding results are shown in Figure 6. We find that the behavior of the flux remains consistent between the two models, with the profiles nearly superimposable. This suggests that the underlying flux dynamics are robust to the choice of the dynamo mechanism.



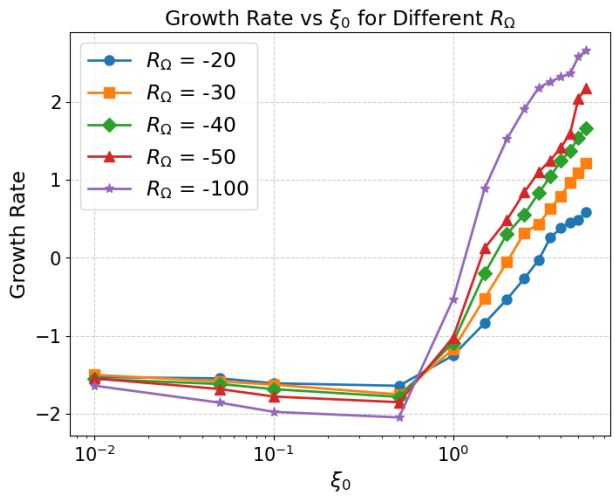
(a) *Mean field strength vs time plots, the y axis here is in logarithmic scale.*

(b) *Mean field strength vs z plots, dashed line represents $\overline{B}_\phi$ and solid lines are $\overline{B}_r$.*

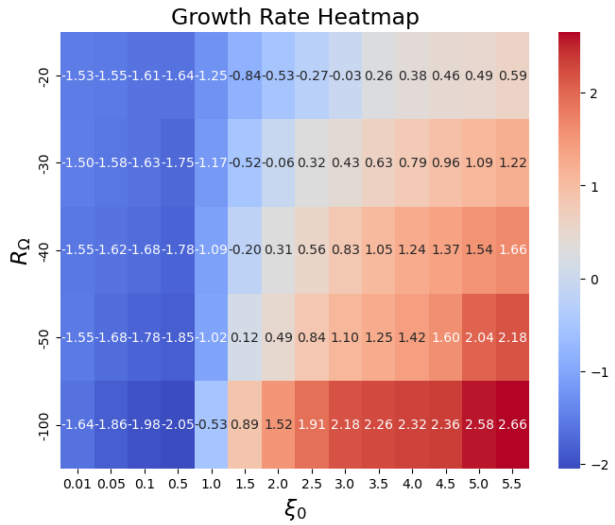
(c) *Total field strength vs z plots. This is the vector sum of $\overline{B}_\phi$ and $\overline{B}_r$.*

Fig. 6: These models were done to test the impact of new flux terms in $\alpha\Omega$ versus α^2-Ω dynamo regimes. All models exclude the kinetic α-effect ($\alpha_k = 0$) and use $\xi_0 = 3.5$. Case 1 and Case 2 correspond to the different velocity profiles given in 15

4.5 Parameter space study



(a) Here the v_0 is set to be a constant of 20 km/s. R_Ω is varied from -20 to -100.



(b) The contour plot representing growth rate changing with respect to ξ_0 and R_Ω

Fig. 7: Plots illustrating the dependence of growth rate on ξ_0 and $R\Omega$. The models do not contain α_k effect here.

We conducted a parameter space study to understand how the growth rate varies with the strength of turbulence and differential rotation. In Figure 7a, we fix $v_0 = 20$ km/s and vary R_Ω from -10 to -80. Our results show that changes in v_0 do not significantly impact the growth rate. However, increasing the magnitude of R_Ω leads to a notable enhancement in the growth rate. This implies that for sufficiently large values of $|R_\Omega|$, a purely Vishniac flux-driven dynamo is possible even for ξ_0 values less than 3.5.

5 The no-z approxmation

To simplify the one-dimensional dynamo equations given in equations 30,31 and 32 in the vertical direction, we adopt the no-z approximation. This method replaces vertical derivatives with algebraic terms:

$$\frac{\partial}{\partial z} \to \frac{2}{\pi h}, \quad \frac{\partial^2}{\partial z^2} \to -\frac{\pi^2}{4h^2}$$

The signs are chosen to reflect boundary conditions and h is the disc semi-thickness, the constants used are from [5]. In the steady state, this reduces the system to a set of algebraic equations, effectively representing vertically averaged fields. For the kinematic regime, where magnetic fields grow exponentially like $\mathbf{B}(t) \propto e^{\gamma t}$, the approximation allows us to solve for the growth rate γ directly.

Under this approximation, the mean field evolution equations become:

$$\gamma B_r = -\frac{2}{\pi h}(\alpha B_\varphi) - \frac{\eta \pi^2}{h^2} B_r \tag{42}$$

$$\gamma B_\varphi = -\Omega B_r - \frac{\eta\pi^2}{4h^2} B_\varphi \tag{43}$$

From equation 42, solve for B_r and substitute it in equation 43 to get:

$$\gamma = \pm\sqrt{\frac{2\Omega\pi\alpha}{h}} - \frac{\eta\pi^2}{4h^2} \tag{44}$$

Now, we are setting the α_k to 0 to study how the Vishniac flux is alone driving the dynamo, thus $\alpha = \alpha_m$:

$$\gamma = \pm\sqrt{\frac{2\Omega\pi\alpha_m}{h}} - \frac{\eta\pi^2}{4h^2} \tag{45}$$

The first term is the driving term and the second is the diffusion term. We are interested in growing solutions, so we continue with the positive sign for the first term.
Now, to find the exact value of α_m for which growth starts, we can do the simple case where $\gamma = 0$ and solve for α_m. We get:

$$\alpha_m > \frac{\eta^2\pi^3}{32h^3\Omega}. \tag{46}$$

Now we solve the equation 32 using no-z. During the initial kinematic regime, generated mean fields are not significant in the disc, so the first term of equation 32 goes to zero and we are left with the divergence of Vishniac and diffusive fluxes,

$$\frac{\partial\alpha_m}{\partial t} = -\nabla\cdot\boldsymbol{\mathcal{F}^D} - \nabla\cdot\boldsymbol{\mathcal{F}^{NV}}$$

Evidently, there is no growth of α_m, if there is no α_k too to carry the α effect, there will not be a magnetic field growth. But let us look at the form of these helicity fluxes using the no-z simplification.
The no-z form of Case 1 Vishniac flux is:

$$-\nabla\cdot F^{NV} = \Omega\frac{\sqrt{2}\tau^2 v^2 b^2}{h}\left(\frac{45}{16200}\xi_0 - \frac{2503}{16200}\right)$$

This means there is a critical value ξ_{crit} above which the flux flips sign, becomes helicity-ejecting and aids the dynamo action, it is given by:

$$\xi_{\text{crit}} = \frac{1285}{2520} \approx 0.506.$$

Therefore:

- For $\xi_0 > 0.506$, the flux promotes helicity ejection, supporting dynamo action.
- For $\xi_0 < 0.506$, the flux suppresses helicity transport, leading to resistive quenching.

Similarly, the no-z form of Case 2 Vishniac flux is:

$$-\nabla\cdot F^{NV}\Big|_{\text{no-}z} = \Omega\,\frac{\sqrt{2}\,\tau^2\,v^2 b^2}{h}\left(\frac{7}{45}\xi_0 - \frac{2503}{16200}\right).$$

The critical value ξ_{crit} above which the flux becomes helicity-ejecting in this case is 0.993.

Doing the same approximation on diffusive flux with Case 1 and case 2 turbulent velocity

profile, at the midplane, we will get the same expression:

$$-\nabla \cdot \boldsymbol{\mathcal{F}}^{D}\Big|_{\text{no-z}} = -\frac{7\pi^2\tau v_0^2(1+\xi_0)}{27h^2}h^b \tag{47}$$

Now, going back to α_m evolution, we will have:

$$0 = -\nabla \cdot \boldsymbol{\mathcal{F}}^{\boldsymbol{D}} - \nabla \cdot \boldsymbol{\mathcal{F}}^{\boldsymbol{NV}}$$

Substitute fluxes; since both cases are similar except for the constants in the Vishniac flux, we will do Case 1 here:

$$\Omega\frac{4\pi\rho\sqrt{2}\tau^2 v^4\xi_0}{h}\left(\frac{7}{45}\xi_0 - \frac{2503}{16200}\right) = \frac{7\pi^2\tau v_0^2(1+\xi_0)}{27h^2}h^b.$$

Assume $h^b \propto \alpha_m$ (helicity-α_m closure) and equate fluxes:

$$\alpha_m \propto \frac{108\pi\rho\sqrt{2}\Omega\tau^2 v^4 h\xi_0}{7\pi^2(1+\xi_0)v_0^2}\left(\frac{7}{45}\xi_0 - \frac{2503}{16200}\right).$$

At Large ξ_0, i.e., magnetically dominated turbulence the Vishniac flux dominates (positive divergence), the first term of α_m very higher than the second. This means we have an α_m contribution to total α effect from the initial time of dynamo action. High ξ_0 means strong small-scale fields ($b^2 \gg v^2$), but the Vishniac flux itself is a solution to this problem because it is a helicity flux. Thus, the system self-regulates and give rise to a dynamo effect at super-equipartition ξ_0 values.

6 Conclusions

In this study, we introduced the Vishniac flux into numerical dynamo models to assess its role in the evolution of galactic magnetic fields. Our primary question was whether this new flux could significantly alter our understanding of mean-field dynamo theory. The short answer is, not by much. The Vishniac flux alone does not independently drive a dynamo to saturation for realistic values of ξ_0.

Next, we explored whether adding this flux to a conventional dynamo, where α_k drives the system, would enhance or hinder magnetic field growth. The answer depends on the circumstances. The flux expression introduces a saddle point where it flips sign, and this transition occurs at different values of ξ_0 depending on the turbulence profile. For turbulence peaking at the disc edges, the sign flip occurs around $\xi_0 = 0.506$, and for turbulence peaking at the midplane, it's closer to 1. Below these values, the Vishniac flux suppresses dynamo action, while above them, it enhances it. However, the effect is not particularly dramatic.

When examining very high ξ_0 values, the flux becomes significantly stronger due to its quadratic dependence on ξ_0, and under these conditions, the Vishniac flux alone can drive dynamo action. While this scenario is not realistically applicable to typical galactic systems, it is theoretically intriguing. It represents a form of magnetically-driven dynamo action, a departure from the conventional kinetically-driven models, adding a new layer of complexity to our understanding of dynamo processes.

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